

Compositionality in the depth-charge illusion: Some experimental insights^{*}

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Abstract. I present an account of the depth-charge illusion based on presupposition accommodation, which makes truth-conditional predictions that diverge from two recent analyses. This is supported by experimental evidence employing a novel setup with an abstract, visual context that minimises interference from real-world biases.

Keywords: depth-charge illusion · presupposition accommodation · degree semantics

1 Introduction

The phenomenon now known as the ‘depth-charge illusion’ is best exemplified by comparing (1) and (2) [18]. Taking (1) to mean “ban all missiles, however small”, a parallel decomposition of (2) is expected to have a suspicious meaning: “Ignore all head injuries, however trivial”. A series of experiments has established, however, that (2) receives high acceptability/sensicality ratings and is not read with this meaning, thereby leading to the label of an illusion [5, 10, 15, 18].

- (1) No missile is too small to be banned.
- (2) No head injury is too trivial to be ignored.

Following an overview of the formal components required to analyse the sentences in question and a summary of two recent analyses, I develop an alternate account of the depth-charge illusion. Finally, I discuss an experiment using a novel setup with an abstract, visual context that minimises interference from real-world biases, the results of which are in line with the new proposal.

2 Background

I employ a framework where degrees are conceptualised as extents on a dimensional scale. Given that a scale is an ordered set of values $\langle 0, \infty \rangle$, extents can

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be defined as the dense subsets of this scale. Positive polar adjectives map to positive extents of the form $\langle 0, n \rangle$, while negative polar adjectives like those in (1) and (2) map to negative extents [hereafter e^-] of the form $\langle n, \infty \rangle$ where n is the output of a measure function [11, 9, 16].

The denotation for a negative polar adjective would then be as in (3), returning TRUE for an interval delimited on the *lower* end by the height of x or any point value on the scale that x **does not** reach.

$$(3) \quad \llbracket \text{small} \rrbracket = [\lambda e \in D_d. [\lambda x \in D_e. e \text{ is a negative extent of the form } \langle n, \infty \rangle \\ \& \text{HEIGHT}(x) \leq n]]^1$$

The degree operator ‘*too*’ is analysed as in (4) following [11]. It relates a world argument, the modalised predicate as an incomplete conditional (*the e' such that if the missile is e' -small, it can be banned*), and an extent predicate (*the e such that the missile is e -small*). A simplified version of [11]’s definition of *can*, which is taken to be the covert modal in sentences like (1), is given in (5). The phrase w' ACC w in (5) abbreviates $\llbracket \text{ACC} \rrbracket(w)(w')$, where $\llbracket \text{ACC} \rrbracket$ is defined as in (5a).

$$(4) \quad \llbracket \text{too} \rrbracket = [\lambda w \in D_s. [\lambda P \in D_{\langle s, \langle \langle s, t \rangle, t \rangle \rangle}. [\lambda E \in D_{\langle d, \langle s, t \rangle \rangle}. \\ \text{MAX}(\lambda e. E(e)(w)) > \text{MAX}(\lambda e'. P(w)(E(e')))]]]]$$

$$(5) \quad \llbracket \text{can} \rrbracket = [\lambda w \in D_s. [\lambda p \in D_{\langle s, t \rangle}. [\lambda q \in D_{\langle s, t \rangle}. \\ \exists w' \text{ ACC } w \& p(w') = 1 \& q(w') = 1]]]$$

$$\text{a. } \llbracket \text{ACC} \rrbracket = [\lambda w \in D_s. [\lambda w' \in D_s. \\ w \text{ is accessible from } w \text{ if } w' \text{ is compatible with what is known in } w]]$$

An extent being greater than another is defined in terms of subset relations (6a) [9, 11]. Based on (6a), the MAX extent can be selected as in (6b):

$$(6) \quad \text{a. } \forall e_1, e_2 \in D_d : e_1 > e_2 \text{ iff } e_1, e_2 \text{ are of the same polarity } \& e_1 \cap e_2 = e_2 \& e_1 \neq e_2 \\ \text{b. } \text{MAX}(E) = \iota e \in E. \forall e' \in E : e > e' \vee e = e'$$

These elements merge into the following truth conditions for (1), abbreviated hereafter as (7b).

$$(7) \quad \text{a. } \llbracket (1) \rrbracket = 1 \text{ iff} \\ \neg \exists x. x \text{ is a missile in } @ \& \text{MAX}([\lambda e. x \text{ is } e\text{-small in } @]) > \\ \text{MAX}([\lambda e'. \exists w' \text{ ACC } @ \& x \text{ is banned in } w' \& x \text{ is } e'\text{-small in } w']) \\ \text{there is no } x \text{ such that } x \text{ is a missile in } @ \text{ and the maximal } e \text{ such that} \\ x \text{ is } e\text{-small in } @ \text{ is greater than the maximal } e' \text{ such that, according} \\ \text{to what is known in } @, \text{ if } x \text{ is } e'\text{-small, } x \text{ is banned.}$$

¹ Extents may be referred to as being, e.g., ‘ e^- -small’ in the text. This is merely shorthand to indicate a negative height extent, not intended to invoke a scale of *smallness*.

b. $\neg\exists$ missile: MAX size e^- of missile $>$ MAX size e^- banned

(1) and (2) are otherwise parallel, and both compare negative extents. The key difference is in the monotonicity of the standard of comparison [7]. The interpretation context of (1) is such that *less* small (=larger) missiles are more likely to be banned [Fig. 1]. Following the framework in [3], this is an *upward scalar* context: if a missile of size x is banned, it is entailed that missiles larger than x are also banned. Conversely, the context for (2) and (11) is *downward scalar*: *more* trivial injuries are generally more likely to be ignored [Fig. 2].

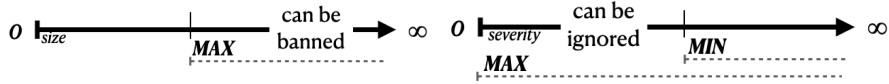


Fig. 1: Upward scalarity in (1)

Fig. 2: Downward scalarity in (2) & (11)

Downward scalarity is common to ‘illusion’ examples with negative polar adjectives. It has the effect that the MAX extent of the standard of comparison (*the MAX e^- such the head injury can be ignored if it is e^- -trivial*) spans over all points on the scale, meaning that no extent can be greater. (2) would thus indeed not be interpretable with truth conditions parallel to (7), given in (8):

(8) $\neg\exists$ head injury: MAX severity e^- of injury $>$ MAX severity e^- ignored

2.1 Previous approaches

i. Maximal Informativity (hereafter: **Max-Inf**) This approach suggests that *too* compares the subject to the *maximally informative* extent of the standard of comparison. For (2), downward scalarity makes the **minimal** e^- that *can* be ignored more informative than the maximal when negative extents are compared. This generates the truth conditions in (9), where (2) is TRUE *iff* there is no head injury that can be ignored.

(9) $\neg\exists$ head injury: MAX severity e^- of injury $>$ MIN severity e^- ignored

These conditions make the sentence interpretable in a way that explains its sensicality in the context of the real world, given that head injuries are generally *not* to be ignored. But, (9) overgenerates in examples without the initial negation. Example (8) below illustrates this with a downward scalar context parallel to (2). Truth conditions as per (9) predict that (8) is acceptable in the given context, contrary to intuition.

Context: On some highways, speeds lower than 45km/h are fined.

John is driving at 35 km/h.

(10) # John is too slow to be fined.

This account would thus predict that (11) is also acceptable. This has in fact not been attested experimentally: deleting the sentence-initial negation dials back the ‘illusion’ effect, such that sentences like (11) tend *not* to be considered sensical, unlike (2) [15].

(11) # Some head injuries are too trivial to be ignored.

ii. Hypernegation (hereafter: **HyperNeg**)² This processing-based approach posits that the increase in processing load from the sentence-initial negation causes an error in the composition otherwise proceeding as given in (8). This error causes *hypernegation*, such that the sentence is interpreted following the truth conditions in (12). These compare the subject to the *upward scalar* counterpart of the standard of comparison, i.e., to extents that *cannot* be ignored. The non-interpretability faced under downward scalarity is thus avoided.

(12) $\neg \exists$ head injury: MAX sev. e^- of injury > MAX sev. e^- not ignored

These conditions are aligned with those from the **Max-Inf** approach (9), returning TRUE *iff* there is no head injury that can be ignored. However, since the negation (and its associated processing load) is responsible for the error that derives (12) from (8), this account does not overgenerate in sentences without the negation, like (11). The low acceptability of (11) is explained as arising due to the context not being upward scalar so as to be interpretable under (8).

3 Proposal

I take the data point illustrated by (11) as a crucial clue to this puzzle. Conceptually, the proposal is as follows: An injury being *too trivial to be ignored* requires there to be a lower threshold to what injuries can be ignored. Contextually, there is no such lower threshold available that is distinct to the scale minima. Modelling this requirement as a **presupposition** triggered by *too* makes the non-negated version in (11) a simple case of presupposition failure. The ‘illusion’ version (2) can then be analysed as an instance of local accommodation under negation³[6, 8]. I implement this formally below.

² For brevity, I only discuss one prominent approach among those [13, 14] positing a processing-driven adjustment to an otherwise compositional parse .

³ This can alternatively be modelled as *metalinguistic negation*[8, 12], as was done in the initial version of this paper. However, as noted by reviewers, this is difficult to bolster with independent evidence. For instance, NPIs are allowed: *No head injury is too trivial to ever be ignored*. But, *too* appears to license NPIs—*John is too rich to ever get into trouble with the law*—rendering a standard test of metalinguistic negation unreliable. While it can be argued that this does not, then, constitute evidence *against* metalinguistic negation, the accommodation analysis developed here can model the same intuitions parsimoniously.

Step 1 : Take *too* to presuppose that the maximal extent satisfying the standard of comparison is a proper subset of the scale. The truth conditions of *too* can then be updated as follows(13).

$$(13) \quad \llbracket too \rrbracket = [\lambda w \in D_s. [\lambda P \in D_{\langle s, \langle \langle s, t \rangle, t \rangle \rangle}. [\lambda E \in D_{\langle d, \langle s, t \rangle \rangle}. \text{MAX}(\lambda e. E(e)(w)) > \text{MAX}(\lambda e'. P(w)(E(e')))]]]]$$

$\llbracket too \rrbracket$ is only defined if the $\text{MAX}(\lambda e'. P(w)(E(e')))$ is not of the form $\langle 0, \infty \rangle$.

The intuition behind the above formulation is this: under *too*, the comparee is required to map to a MAX extent greater than the MAX extent of the standard of comparison. If the latter extent spans the entire scale, there is no extent greater than this that could *in principle* be MAX extent corresponding to the comparee.

When comparing negative extents, this presupposition would not be met whenever the context is downward scalar, as in (2) and (11) [recall Fig. 2]. Since the MAX e^- *such the head injury can be ignored if it is e^- -trivial* spans the entire scale, explaining the low acceptability of (11) .

Step 2 : Interpret (2) via local accommodation under negation [6, 17]. The intuition for this step is well illustrated by example (14). Since John has not bought a car, it is (trivially) true that he also does not *regret* buying one.

Context: John has not bought a car.

$$(14) \quad \text{John does not regret buying a car.}$$

A similar effect can be seen in (2), where, as discussed above, the MAX e^- *such the head injury can be ignored if it is e^- -trivial* spans over all points on the scale [Fig. 2]. Therefore, since no head injury can—even in principle—have a greater e^- than this, it is (trivially) true that no such injury *exists*.

This can be modelled in dynamic semantics through the process of **accommodation** [6]. In this framework, if a presupposition is not already part of the context, it can often be *globally accommodated*. Informally, this means that we adjust the context to one where the presupposition *is* met, and then proceed to add the (in this case, negated) sentence to it.

However, this is not always possible, particularly if the presupposition to be accommodated contradicts the content of the context set. In the case of sentence (14), the proposition that John has *not* bought a car being in the context c would forbid us from globally accommodating the presupposition p that John *did* buy a car. In this case, we have the option of accommodating the presupposition p **locally** in order to interpret the sentence S , as defined in (15). I propose that (2) may be interpreted similarly, given that the presupposition in (13) is not viable at all given the downward scalar context.

$$(15) \quad c + \neg S = c - ((c + p) + S)$$

Local accommodation is standardly modelled through the intervention of the assertoric ‘floating *A*’ operator [2]. When this operator takes a proposition that is *neither TRUE nor FALSE* in a trivalent logic, it is mapped to FALSE. Under negation, the LF would then be as in (16), returning TRUE.

$$(16) \quad \neg[A[\exists \text{ head injury: MAX severity } e^- \text{ of injury} > \text{MAX severity } e^- \text{ ignored}]]$$

Summary The claim is that *too* encodes a presupposition requiring the existence of some interval on the scale that is greater than the MAX extent of the standard of comparison (here, the MAX extent that can be ignored). When this is not met, a *negated* version of the sentence can still be true through local accommodation, with the following meaning: *No head injury is too trivial to be ignored, because no head injury can be more trivial than what can be ignored*. Our proposal (hereafter: **LocAcc**) thus provides an integrated explanation for both, the sensicality of (2) and the incoherence of (11). Furthermore, it generalises to ‘illusion’ examples with positive polar adjectives as well, such as (17):

$$(17) \quad \text{No mountain is too high to stop us}(\dots). \quad [4]$$

When comparing positive extents, the proposed presupposition in (13) would not be met under *upward* scalarity, which (17) possesses: *higher* mountains are more likely to stop us; there is no *upper* threshold to heights that can be an obstacle. Once again, the MAX e^- *such the mountain can stop us if it is e^- -high* would span the entire scale. This would then trigger presupposition failure in the absence of the negation, and allow (17) to be interpretable as TRUE through local accommodation.

4 Experimental Data

I present data from a truth value judgement task⁴ with diverging predictions for the approaches discussed. An abstract version of the target sentences was used (18) with visual judgement contexts, to avoid interaction with world knowledge.

$$(18) \quad \text{The/No opening is too small to be blocked.}$$

These sentences were first tested with images enforcing a non-scalar context for what can be blocked. As expected, responses aligned with truth conditions in (8); i.e., there was no illusion effect. This affirmed the role of scalarity in driving the phenomenon. The experiment presented here reflects the current puzzle,

⁴ Despite the proposal involving presupposition failure, the task reported here allowed only bivalent TRUE/FALSE judgements. The decision against offering a *can’t say/neither* button was due to a concern that the complexity of these sentences may lead to it being overused. It has, however, been independently been observed that *judgements* under presupposition failure often default to FALSE (or TRUE under negation), even in the presence of a third option[1].

implementing a **downward scalar** context matching that of (2) & (11): the setup is such that all openings smaller than a certain size can be blocked by the brick [see Fig 3].

400 one-shot responses across 4 conditions (2x2) were collected, crossing SENTENCE TYPE (negated vs. definite sentence), with OPENING SIZE (openings in the image are *bigger* than the brick/cannot be blocked; vs. *smaller*/can be blocked). Instructions prior to the experimental trial illustrated that an opening counts as blocked if its breadth is covered by the brick, as seen in the *smaller* conditions below.

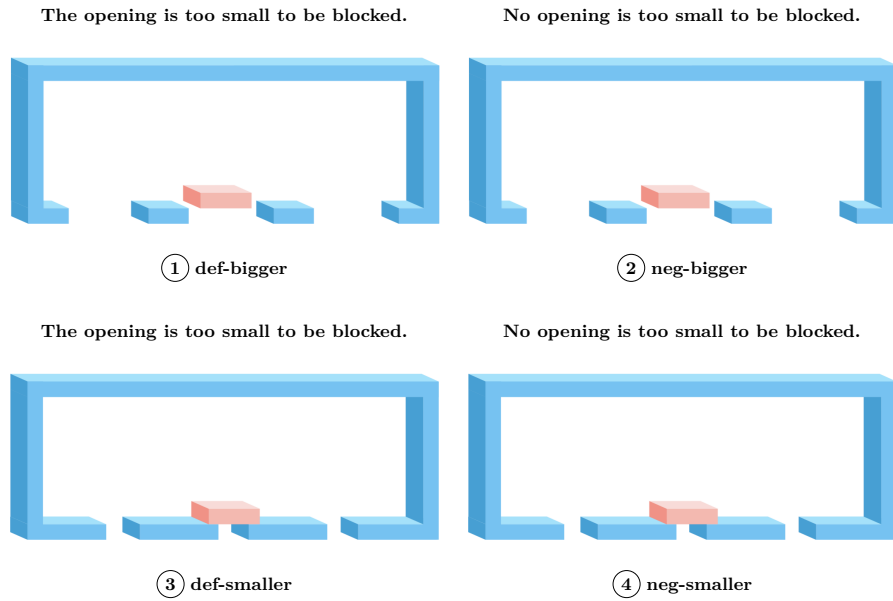


Fig. 3: Experimental items

4.1 Predictions

As shown in Table 1, the predicted response pattern varies for each account.

Table 1: Predictions from different approaches

Account	Conditions			
	(1) def-bigger	(2) neg-bigger	(3) def-smaller	(4) neg-smaller
<i>Max-Inf</i>	F	T	T	F
<i>HyperNeg</i>	?/ F	T	?/ F	F
<i>LocAcc</i>	?/ F	T	?/ F	T

Definite sentences : Both the *HyperNeg* and the *LocAcc* approaches predict FALSE/unsystematic responses for (1) & (3), in line with presupposition failure. The *Max-Inf* approach expects a TRUE response as long as the opening is smaller than the *MIN* e^- that can be blocked ((3))⁵ and FALSE otherwise ((1)).

Negated sentences : (2) is predicted by all approaches to be TRUE since it mirrors the real world context for (2). Specifically, head injuries in the real world are presumed to be *more severe* (less trivial) than what can be ignored. Similarly, openings in this context are *bigger* (less small) than what can be blocked.

The crucial case is (4), where both previous accounts diverge from the *LocAcc* account. Recall that the *Max-Inf* and *HyperNeg* accounts posit adapted truth conditions in (9) and (12) such that the negated sentence is true *iff* there exists no head injury that can be ignored. In (4), however, these truth conditions are not met: there *do* exist openings that are can be blocked. Both previous accounts thus predict that a FALSE response.

By contrast, the *LocAcc* negation account predicts that (2) is TRUE even if there exists some head injury that can be ignored, since there is still no injury *too trivial* to be ignored. Hence, (4) is also predicted to be true.

4.2 Results & Discussion

The response pattern, plotted in Fig. 4, shows alignment with the predictions for the *LocAcc* account. The *Max-Inf* approach fails to capture the tendencies for (3) and (4), while *HyperNeg* does not explain (4). It appears that a sentence like (2) is indeed considered TRUE even when the judgement context does not fulfil the truth conditions in (9) or (12).

The responses were analysed with a GLM modelling the log odds of a TRUE response as: $\text{RESPONSE} \sim \text{SENTENCE TYPE} * \text{OPENING SIZE}$. With *def* and *bigger* as reference levels (*Int.Est.*: -1.7346, *SE.*: 0.2801, *z.*: -6.194), there was a significant main effect of SENTENCE TYPE (*Est.*: 2.6791, *SE.*: 0.3578, *z.*: 7.487) followed by a smaller effects of the interaction (*Est.*: -1.2475, *SE.*: 0.4692, *z.*: -2.659) and OPENING SIZE (*Est.*: 0.7763, *SE.*: 0.3576, *z.*: 2.171). Model comparison through an analysis of variance (ANOVA) is shown in Table 3, referring to the GLMs in Table 2. In line with the predictions for the *LocAcc* account, the best fit is from a model with only SENTENCE TYPE as predictor.

The data also presents some additional questions. Firstly, the model comparison suggests that a GLM with both factors and an interaction term also provides a good fit. However, as is apparent in Fig. 4, this interaction does not flip the majority tendency: rather, it appears to come from an increase in variability within the *smaller* conditions compared to the *bigger* ones. If presupposition

⁵ In other words, the *Max-Inf* account looks for an opening *small enough* to be blocked: if it exists, the definite sentence will be TRUE and the negated one would be FALSE. In *HyperNeg*, only the latter applies.

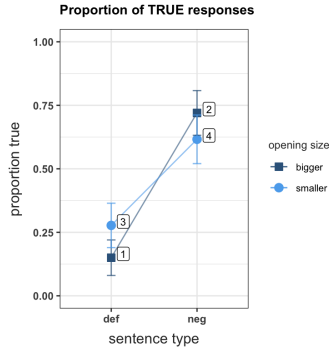


Fig. 4: Responses

Table 2: GLMs compared in ANOVA

#	Model
a.	response ~ 1
b.	response ~ SENTENCE TYPE
c.	response ~ SENTENCE TYPE+OPENING SIZE
d.	response ~ SENTENCE TYPE*OPENING SIZE

Table 3: ANOVA results

#	Res. DF	Res. Dev.	DF	Dev.	$Pr(< z)$
a.	399	548.74			
b.	398	461.55	1	87.190	$< 2.2e-16^{***}$
c.	397	461.48	1	7.249	0.780948
d.	396	454.23	1	7.249	0.007093**

failure indeed plays a role in judgements for the *smaller* conditions, this would potentially be explained by the expected increase in cognitive effort. Secondly, the preference for a response emerges less strongly in the *neg* conditions than in the *def* ones. This may also be due to the increase in the processing load under negation, which has been attested independently [15], regardless of whether it is considered primarily responsible for the ‘illusion’.

Such an attenuation in the strength of the results under negation was also observed for unrelated data on presupposition failure [1], suggesting that these features of the data may not be specific to the phenomenon being discussed here. If this is the case, however, more work is needed in order to understand how the variability is distributed between these influences.

5 Conclusion

This paper proposes a novel account of the depth-charge illusion by positing a presupposition for *too* and suggesting that negated sentence (2) is then interpreted via local accommodation. It also presents data from an experiment that makes diverging predictions for the current account and two recent proposals, with results that are in line with the presuppositional approach.

This proposal would entail, akin to the **Max-Inf** account, that there is nothing problematic with depth-charge *configuration* itself: rather, it is the scalarity of the standard of comparison—which comes from the context or from world knowledge—that sets up impossible comparisons in interaction with other features of the sentence, adjectival polarity in particular.

A presuppositional account would also predict a preference for *global* accommodation, when this is not hindered. At first glance, this appears to be borne out: changing (1) to *No missile is too large to be banned* does not necessarily cause it to behave like an ‘illusion’ example. Instead, we seem to get a *Ban all missiles*

reading—globally accommodating the presupposition that large missiles are less likely to be banned in the utterance context. I leave the investigation of these predictions to future work.

References

1. Abrusan, M., Szendroi, K.: Experimenting with the king of France: Topics, verifiability and definite descriptions. *Semantics and Pragmatics* **6** (Oct 2013). <https://doi.org/10.3765/sp.6.10>
2. Beaver, D., Krahmer, E.: A Partial Account of Presupposition Projection. *Journal of Logic, Language and Information* **10**, 147–182 (2001)
3. Beck, S.: Lucinda Driving Too Fast Again—The Scalar Properties of Ambiguous Than-Clauses. *Journal of Semantics* **30**(1), 1–63 (Feb 2013). <https://doi.org/10.1093/jos/ffr011>
4. Fortuin, E.: Deconstructing a verbal illusion: The ‘No X is too Y to Z’ construction and the rhetoric of negation. *Cognitive Linguistics* **25**(2), 249–292 (Jun 2014). <https://doi.org/10.1515/cog-2014-0014>
5. Giannouli, V.: A Verbal Illusion Reexamined. *Acta Neuropsychologica* **14**(4), 323–329 (Oct 2016). <https://doi.org/10.5604/17307503.1225267>
6. Heim, I.: On the Projection Problem for Presuppositions. *Proceedings of the Second West Coast Conference on Formal Linguistics* pp. 114–125 (1983). <https://doi.org/10.1002/9780470758335.ch10>
7. Hohaus, V., Bade, N.: When Maximality Is Not Maximally Informative: A Compositional Analysis of the Depth-Charge Illusion. *Proceedings of the 23rd Amsterdam Colloquium* pp. 113–119 (2022)
8. Horn, L.R.: Metalinguistic Negation and Pragmatic Ambiguity. *Language* **61**(1), 121–174 (Mar 1985). <https://doi.org/10.2307/413423>
9. Kennedy, C.: Polar Opposition and the Ontology of ‘Degrees’ (2001). <https://doi.org/10.1023/A:1005668525906>
10. Kizach, J., Christensen, K.R., Weed, E.: A Verbal Illusion: Now in Three Languages. *Journal of Psycholinguistic Research* **45**(3), 753–768 (Jun 2016). <https://doi.org/10.1007/s10936-015-9370-6>
11. Meier, C.: The Meaning of Too, Enough, and So . . . That. *Natural Language Semantics* **11**, 69–107 (2003). <https://doi.org/10.1023/A:1023002608785>
12. Moeschler, J.: Negation, scope and the descriptive/metalinguistic distinction. *Generative Grammar in Geneva* **6**, 29–48 (2010)
13. O’Connor, E.: Comparative Illusions at the Syntax-Semantics Interface. PhD Thesis, University of Southern California (2015)
14. O’Connor, E.: The Accidental Ambiguity of Inversion Illusions. In: Lamont, A., Tetzloff, K. (eds.) *Proceedings of NELS*. vol. 47, pp. 329–342. Graduate Linguistics Student Association, University of Massachusetts, Amherst, MA (2017)
15. Paape, D., Vasishth, S., Von Der Malsburg, T.: Quadruplex Negatio Invertit? The On-Line Processing of Depth Charge Sentences. *Journal of Semantics* **37**(4), 509–555 (Nov 2020). <https://doi.org/10.1093/jos/ffaa009>
16. von Stechow, A.: Comparing Semantic Theories of Comparison. *Journal of Semantics* **3**(1-2), 1–77 (1984). <https://doi.org/10.1093/jos/3.1-2.1>
17. Von Steinhilber, K.: What is presupposition accommodation, again? *Philosophical Perspectives* **22**(1), 137–170 (Dec 2008). <https://doi.org/10.1111/j.1520-8583.2008.00144.x>
18. Wason, P.C., Reich, S.S.: A Verbal Illusion. *Quarterly Journal of Experimental Psychology* **31**(4), 591–597 (Nov 1979). <https://doi.org/10.1080/14640747908400750>